

2017-Ph-H1-Q22

Section: Particles and Waves

Topic: Refraction of Light

A ray of light passes from air into glass.

Angle of incidence and angle of refraction are given.

Find the refractive index of the glass.



Use Snell's Law:

$$n = \frac{\sin i}{\sin r}$$

Given:

- $i = 60^\circ$,
- $r = 36^\circ$

$$n = \frac{\sin 60^\circ}{\sin 36^\circ} = \frac{0.866}{0.588} \approx 1.47$$



Final Answer: C — 1.47



Revision Tips:

- Snell's Law:

$$n_1 \sin i = n_2 \sin r$$

- Refractive index > 1 means **light slows down**